

Avid Mbox Pro on modern macOS

Installing the ASFireWire driver on a new machine — step by step. Written 30 September 2026.

What this is

The Avid Mbox Pro (3rd generation, 2011) is a FireWire audio interface. Avid's own driver stopped at macOS Mojave, and macOS Tahoe (26) removed Apple's FireWire stack altogether. **ASFireWire** rebuilds that stack in userspace with DriverKit, and since 30 September 2026 it supports the Mbox Pro directly.

When this works you get all eight channels at once: six line outputs, S/PDIF, both headphone pairs and the four XLR inputs.

Before you start

Requirement	Notes
macOS 26 (Tahoe) or later	Apple Silicon. Earlier macOS still has Apple's own FireWire stack, which claims the hardware.
Xcode 27 or later	Must provide the DriverKit 27 SDK. Xcode 16 will not build this project.
XcodeGen	brew install xcodegen
The cable chain	Mbox Pro → FW400 → FW800 → Thunderbolt 2 → USB-C adapter → Mac
An admin password	Several steps need sudo.

You will have to turn off System Integrity Protection. This build is signed ad-hoc, and the DriverKit entitlements it needs are restricted: macOS validates them against a provisioning profile signed by Apple, and there is no “allow anyway” dialog for that check. Disabling SIP is what makes the system skip it. See the last page for what that means and the alternative.

Step 1 — Disable SIP

Shut down. Hold the **power button** until “Loading startup options” appears. Choose **Options** → **Continue**, pick your volume and authenticate. Then open **Utilities** → **Terminal** and run:

```
csrutil disable
```

Confirm, enter your admin password, then reboot. Back in macOS, verify and enable developer mode:

```
csrutil status
sudo systemextensionsctl developer on
```

The first must report disabled; the second Developer mode set to on.

Step 2 — Point the tools at Xcode 27

```
sudo xcode-select -s /Applications/Xcode.app
sudo xcodebuild -license accept
```

```
sudo xcodebuild -runFirstLaunch
```

Adjust the first path if your Xcode 27 is named differently. Then confirm the SDK is the right one — it must print 27.0 or higher:

```
xcrun --sdk driverkit --show-sdk-version
```

The last two commands matter more than they look. If you copied Xcode from another disk or volume rather than installing it from the App Store, its licence is not accepted and its shared components are missing — they live outside the app bundle, in `/Library/Developer`, and do not travel with a copy. Skipping them produces a build that dies before compiling anything, with an error about a plug-in that does not mention the real cause.

Note those shared components are global, one set per machine, not one per Xcode. If you keep two Xcode versions and the other one starts failing, point `xcode-select` at it and run `runFirstLaunch` again.

Step 3 — Build

```
git clone https://github.com/mrmidi/ASFireWire.git
cd ASFireWire
xcodegen generate
./build.sh --set CODE_SIGNING_ALLOWED=YES \
--set CODE_SIGNING_REQUIRED=YES \
--set CODE_SIGN_IDENTITY=-
```

The signing flags are not optional. Plain `build.sh` compiles without signing on purpose — useful to check the tree builds, useless for installing. `sysext` rejects an unsigned driver extension with error -67062 and a misleading message about apps outside `/Applications`, which has nothing to do with the path.

A tagged release with an `install.sh` is simpler, but v0.3.1 predates Mbox Pro support. Until a later release exists, build from source.

Step 4 — Install

Copy the built app into place and open it:

```
sudo cp -R build/DerivedData/Build/Products/Debug/ASFW.app \
/Applications/
open /Applications/ASFW.app
```

Connect the Mbox Pro and switch it on. In the app, press **Install**. macOS will ask you to authorise it: **System Settings** → **General** → **Login Items & Extensions** → **Driver Extensions**.

Always install from the same place you build from. Testing one copy while building another is the classic way to lose an afternoon.

Step 5 — Verify

```
systemextensionsctl list
system_profiler SPAudioDataType | grep -A6 -i 'mbox'
```

You want **exactly one** entry for `net.mrmidi.ASFW.ASFWDriver`, in state `[activated enabled]`, and the device listed as **Avid Mbox Pro (DICE)** with 8 inputs and 8 outputs at 48 kHz.

One result will look wrong and is not: `system_profiler SPFireWireDataType` returns nothing. That report reads Apple's FireWire stack, which no longer exists. Your bus is now served by a userspace process, invisible to it. Sound while that report is empty is the whole point.

Finally, play something through outputs 1–2. Nothing else proves it.

Three rules that save hours

1. Reboot when you replace the driver with a new version

Swapping a driver extension in place does not work here. The old version hangs in terminating for upgrade via `delegate`, never releases the PCI controller, and the device goes silent — while still appearing in the audio device list. Only a reboot clears it. Expect this every upgrade; it is not a sign anything is broken.

2. Never run `kill` on the driver extension

It kills the process but leaves an orphan entry in `sysextd`. The bus goes empty, `activateDecision` found two entries appears, and even `uninstall` will not complete because the extension is still attached to the PCI device. Reboot instead.

3. Check the list before installing

If `systemextensionsctl list` shows two entries for the driver, reboot *before* trying to install. Installing onto a dirty state does not work and makes the mess worse.

If something goes wrong

Symptom	Cause and fix
Build dies before compiling, plug-in error	Xcode's shared components are stale. <code>sudo xcodebuild -runFirstLaunch</code>
"You have not agreed to the Xcode license agreements"	<code>sudo xcodebuild -license accept</code>
Install fails with -67062	The extension was built unsigned. Rebuild with the three signing flags from Step 3.
Device listed but silent	Usually two extension entries after an upgrade. Check the list, then reboot.
Device does not appear at all	Check the cable chain is connected and the unit is powered. Without hardware the driver does not start, which is correct.
Sound works, controls do not respond	Unplug and replug the FireWire cable. That recovers the control path without disturbing audio. If not, reboot.
Boot panic loop	Boot to Recovery, <code>csrutil disable</code> , boot normally, <code>systemextensionsctl uninstall -net.mrmidi.ASF.ASFWDriver</code>

Outputs 3–4 duplicating 1–2 is not a fault: both pairs are fed from the same mixer bus.

About turning SIP off

SIP is the protection that applies even to the administrator. It cannot be disabled from a running system — not with a password, not with sudo, not as root. It takes physical access, a shutdown, the power button and Recovery. That is precisely its value: it stops software that already has your credentials, not someone standing at your machine.

With SIP off, that guarantee is gone until you turn it back on. Nothing opens to the network and FileVault keeps working, but malicious software you ran and approved could reach places it otherwise could not, and survive attempts to remove it. Only run software you trust, and be suspicious of anything asking for an admin password without a reason you understand.

To restore it, return to Recovery and run:

```
csrutil enable
```

There is a way to keep SIP on. Apple offers development variants of the DriverKit entitlements that need no approval from Apple and work with SIP enabled. They require a paid Apple Developer membership — a free Apple ID cannot obtain a DriverKit provisioning profile. That is the real price: a yearly fee, not paperwork. Two things are untested and would decide it: the development PCI entitlement does not match hardware macOS considers “built in”, and the PCI bridge entitlement has no documented development variant.

ASFireWire is by Aleksandr Shabelnikov (github.com/mrmidi/ASFireWire). Mbox Pro support was contributed in pull request #162 and merged on 30 September 2026. This guide describes a driver signed ad-hoc for personal use; it is not a supported product, and the project is experimental.